Lizard Monitoring System

Temperature, Humidity, and Behavior Tracking

Hank Slaby

Problem Domain

Leopard geckos can live up to **20 years**, but:

- Lifespan is often reduced due to **poor environmental conditions**
- Reptiles are **extremely sensitive** to small changes
- Even a **5°F temperature shift** can impact health
- Owners cannot monitor tanks 24/7
- No direct way for the animal to signal distress

Why This Problem Matters

- Incorrect temperature → digestion issues
- Incorrect humidity → shedding problems
- Behavioral changes are **early warning signals**

Core issue:

> Owners lack continuous, objective monitoring of tank conditions and animal behavior

Solution Overview

A system that:

- Measures **temperature and humidity**
- Tracks **gecko position (hot vs cool side)**
- Combines data into **automated reports**

Technology Stack

- **XIAO ESP32-C6** — control, Wi-Fi, email
- **ESP32-CAM (AI-Thinker)** — vision processing
- **AHT10** — temperature & humidity sensor
- **Arduino IDE** — development environment

Libraries:

- Adafruit_AHT10
- ESP_Mail_Client
- esp_camera

Why This Stack

- Eliminated unnecessary hardware (Arduino Nano)
- ESP32-C6 handles networking + logic
- ESP32-CAM handles vision independently
- Built-in Wi-Fi simplifies email process
- AHT10 provides reliable humidity readings

System Architecture

ESP32-CAM

- Captures frames
- Detects motion location

ESP32-C6

- Reads sensor data
- Receives CAM data (UART)
- Sends email reports

Data Flow:

CAM → UART → C6 → Wi-Fi → Email

How It Works

ESP32-CAM

- Captures grayscale frames
- Splits into left/right regions
- Detects motion

ESP32-C6

- Reads AHT10 every minute
- Receives position data every 10 seconds
- Sends hourly report

Position Detection (Core Algorithm)

- Frame differencing (pixel-by-pixel comparison)
- Detect brightness changes

Classification:

- Left → Cool side
- Right → Hot side
- None → No movement

Email Report Output

- Position summary (% time per side)
- Temperature:
    - Average, min, max
    - °C and °F
- Humidity average
- CAM link status (UP/DOWN)
- Full time-series log

Known Challenges

- Motion threshold tuning
- Camera mounting stability
- Long-term power solution

Learning with AI — Development

- Learned how to structure Arduino code
- Used AI to:
  - Build working sketches
  - Debug errors
  - Select correct libraries

Learning with AI — Hardware Integration

- Wiring ESP32-C6, CAM, and AHT10
- Flashing ESP32-CAM (complex process)

Key takeaway:

> AI accelerated setup, but required validation

Learning with AI — Recovery

- Recovered deleted ESP32-CAM code from chat log
- Save your work!!!

Learning with AI — Limitations

- AI can produce **confidently incorrect answers**
- Small but impactful mistakes occur

Strategy:

- AI → specific questions
- External sources → validation

Demonstration

This demo will show:

- System running live
- Serial communication between boards
- Real-time position tracking
- Email report delivery

Project Links

- Canvas: https://nku.instructure.com/courses/88152/pages/individual-project-hank-slaby
- GitHub: https://github.com/HankA5/CSC494_Project

Questions?